

WASP-10b

R-1240

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_191203 · created 2026-07-31T10:55:22+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458820.6602 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.144 +/- 0.012
Transit depth (Rp/Rs)^2	2.08 +/- 0.36 [%]
Orbital Inclination (inc)	89.94 +/- 0.8
Airmass coefficient 1 (a1)	1.05 +/- 0.019
Airmass coefficient 2 (a2)	-0.0403 +/- 0.0092
Scatter in the residuals of the lightcurve fit is	1.78 %
Best Comparison Star	#3 - [262, 209]
Optimal Method	PSF photometry
Transit Duration (day)	0.093 +/- 0.0017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.144 ± 0.012 vs 0.1592 (deviation 1.05σ)
Duration fit vs published	0.093 d vs 0.0946 d (deviation 0.77σ)
Mid-time O–C	-6.2 min
Depth SNR	10.0
Residual scatter	1.78 %

Light curve

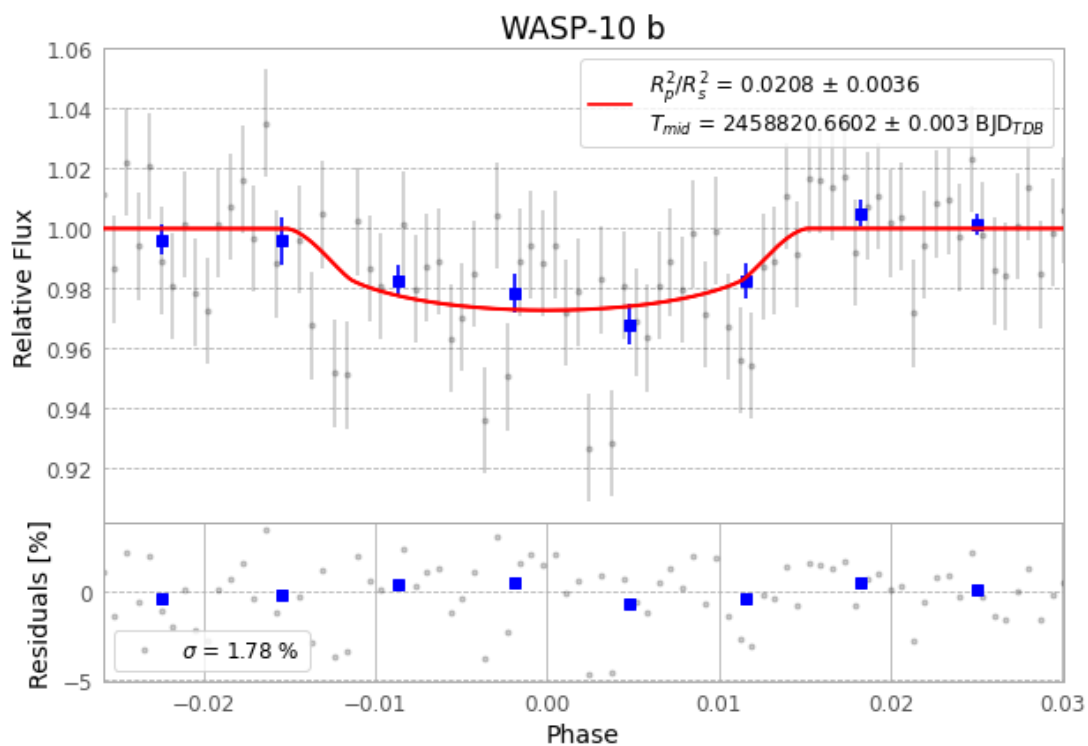


Figure 1 — FinalLightCurve_WASP-10 b_2019-12-02.png (SHA-256 17f3b3a822c234fa...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [265, 308], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 14647a54e18204d5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

84 FITS frames (55 MB), WASP-10191203015112.FITS → WASP-10191203060012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-191203.log (5e3cf2cf0907...), FinalLightCurve_WASP-10 b_2019-12-02.png (17f3b3a822c2...), FinalLightCurve_WASP-10 b_2019-12-02.csv (4c6138f8bc81...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 10:55Z.